PROCESS_DATA : process (clk, initialize) -- sub sensitivity list
variable temp_result : std_logic_vector(2*data_width-1 downto 0); -- temporary variable to store multiplication result
variable temp_data : std_logic_vector(data_width-1 downto 0); -- temporary variable to store data
begin -- process
if initialize = '1' then -- asynchronous reset (active high)
result <= (others => '0');
elsif clk'event and clk = '1' then -- rising clock edge
temp_result := a * b; -- multiplication
temp_data := temp_result(2*data_width-1 downto data_width); -- truncate result to fit output width
result <= temp_data;
end if;
end process;
end behavioral;